MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7030

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Diode
See figure 1
Mounting position: Vertical, anode up or down
Weight: 6 1/4 pounds nominal

ABSOLUTE RATINGS:

Parameter: Unit:	Ef Vac	epx kv	ib a	Ib Adc	Pp kw	tk Sec	TA °C	TE °C	Cooling (Note 1)
MAXIMUM:									
Rectifier:	13.0	25	20	6	2.5	60	45	180	
Charging Diode:	13.75	25	50	2.5	2.5	60	45	180	
TEST CONDITIONS:	13.0	--	--	---	---	--	--	---	

General:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1					
1301	Filament current (1)		If	35	38	Adc
1301	Filament current (2)	Ef = 13.75 Vac	If	36	39.5	Adc
----	Reverse voltage	Ebb = -25 kVdc; t = 5 minutes (Note 3)	--	--	----	---
1256	Electrode current (anode)	Eb = 250 Vdc	Ib	2.9	3.3	Adc
1231	Peak emission	Ef = 13.75 V; eb = 4000v	is	54	---	a
	Quality conformance inspection, part 2					
1032	Vibration, mechanical	Ef = 13.0 Vac±5% Axis-X,Y,Z(see figure 1) Frequency range 10-35 Hz @ .080DA 35-70 Hz @ 5G Sweep time 10-70-10 Hz = 2 min	---	--	No shorts allowed	---
----	Post vibration mechanical end point	Filament current (Note 5)	Δ If	---	1.0	Aac
----	Operation	(Notes 2 and 3)	--	---	---	---
1036	Bump and short	30 degrees hammer drop	-- --	--- ---	--- ---	--- ---
	Quality conformance inspection, part 3					
----	Life test	Group D. (1) Ef = 13.0 Vac; t = 500 hrs (2) Ef = 13.75 Vac; t = 500 hrs.	--	---	---	---
----	Life test end points	If = 13.0 Vac; Ib = Eb = 250 Vdc; is = Ef = 13.75; Eb = 4,000 Vdc (Note 4)	If Ib is	35 2.9 54	38 3.3 ---	Aac Adc a

NOTES:

1. Power dissipation is dependent upon the rate of air flow. The required air flow for various plate dissipation ratings shall be as follows:

Dissipation (kW)	2.5	2.0	1.5	1.0	0.5
Air flow (cfm)	150	120	90	65	50
Pressure (inches of water)	2.5	1.6	.9	.65	.5

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Notes: - Continued

2. A minimum of three tubes shall be operated in a modulator or equivalent circuit for not less than 3 hours per tube. The modulator shall expose the tube to the following conditions:
   epx = 25 kv, ib

3. There shall be no evidence of arcing gas.

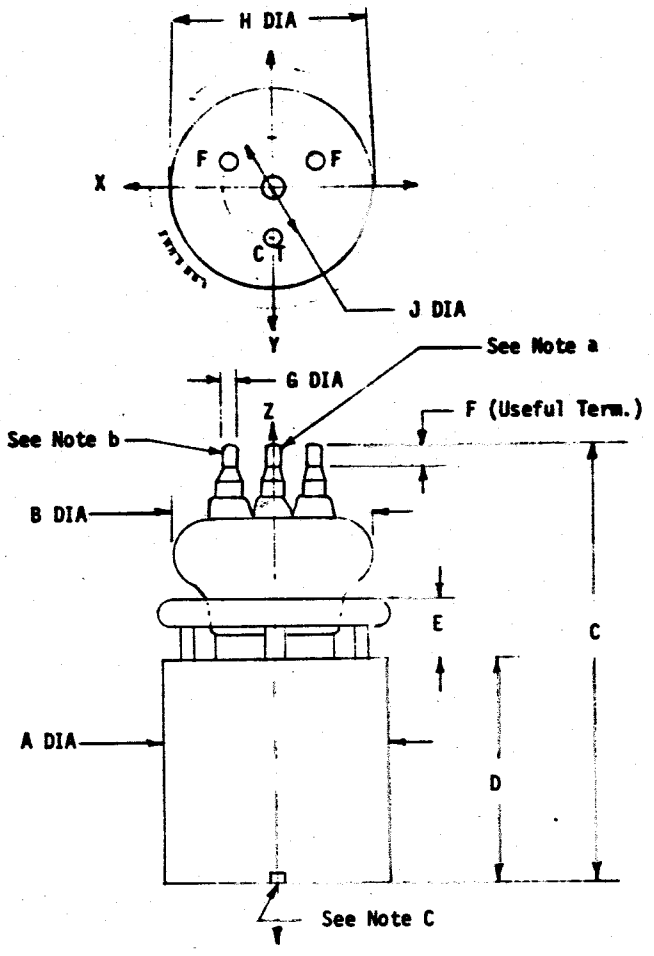
4. Voltage shall be applied in pulses that will not damage the tube.

5. Any change in If resulting from vibration testing shall not exceed the ▲If specified.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity
Navy-EC
(Project 5960-N937)

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LTR	DIMENSIONS	
	Minimum	Maximum
QCI, part 2		
A	3.468	3.532
B		3.250
C	6.500	7.000
D	3.375	3.625
E	.875	.999
F	.375	
G	.360	.390
H		3.250
J	1.375	1.625

NOTES:

a. Tabulation tipoff to be 1/8 inch (min) below terminals.

b. Color code shall be as follows:
Filament terminals - Yellow
Filament centertap - Red

c. Key way to be centered within 5° on same line as filament centertap. Dimensions for key way are .125±.015 X .125±.015 X .062 (min) depth.

Figure 1 - Outline drawing for tube type 7030.

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